Supporting Information A:
BES Data/Code Hack-a-thon Protocol from

Data- and code-archiving in the British Ecological Society journals: present status and recommendations for future improvements.

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Note we will use the BES code of conduct for all in person and online interactions. More details here: https://www.britishecologicalsociety.org/content/code-of-conduct/

Online/remote attendees

Note that the official start date/time of this hack-a-thon is **10am (BST (UTC+1)) 29th September 2025**. Several organisers will attend online only and will facilitate the online component on 29/30th Sept. If you want to work outside these times that's super but please…

1. Do not start data collection in earnest until after **12 noon (BST (UTC+1)) 29th September 2025** so we have time to test and modify the protocol. Feel free to trial and test the protocol on any paper before that, but we will exclude any data collected before this as part of the test data.
2. Check the Discord (and here) for any updates to the protocol.
3. Note any issues you find on the #issues or #options-to-add Discord channels.
4. Make sure to record which papers you are collecting data on by ticking them off in the "done?" column on the BES papers spreadsheet before you start so we don't have too much overlap (not a big deal if it happens but would be good to avoid).

1. Aims

To understand whether data (and code) from published papers in BES journals (plus Ecology and Evolution) are:

- Open and FAIR (https://www.go-fair.org/fair-principles/)
- Being used by others
- Equitably collected and acknowledged

AND…To help attendees see what good (and bad) practice looks like so they can improve their own work.

A few excellent recent resources from SORTEE that you might like to look at before you come:

- https://ecoevorxiv.org/repository/view/9806/
- https://ecoevorxiv.org/repository/view/9948/

2. Getting credit for your input

After you've added data for your first paper, please fill in this form to give us some information about you so we can contact you when we start writing the paper.

Data Recorder Info

Note that just filling this out doesn't guarantee authorship. You will need to confirm later that you have read the paper and agree with how its findings are represented. So please put in an email address that will still be functioning in a year from now.

3. Discord

There is a Discord site for chatting and questions that we will monitor throughout the events on 29/30th September, and less regularly afterwards. You will need to click the invite link in the email and may need to register a Discord account. Let us know if you have any issues.

4. How to collect data

We recommend working in pairs if possible - one person can read the paper and one person can input the data into the Google form. If you're online the Discord channel #find-a-pair can be used to find someone to team up with if you want to.

We have downloaded and cleaned a dataset containing all papers published between 2015 and 2024 (2015 was when the BES data sharing mandate began). Most journals will have data for all ten years, except *People and Nature* and *Ecological Solutions and Evidence* which we started publishing more recently.

The spreadsheet contains summary data on the papers, including their **DOI, a DOI link, the journal, the title of the paper, when it was published** (this is the date for the version of record, not early view), and **the country on the address for the corresponding author**. The papers also have a **paper number**, which we will use to assign papers to data collectors. Note that the order of the papers is randomised to make assigning papers as simple as possible.

4.1 Choosing your papers

The list of papers has been ordered randomly and each given a unique ***paper number***. In person attendees will be given a list of papers to collate data from, online/remote attendees should choose 10 sequential paper numbers.

Please tick ✅ the papers you are collecting data for on the "done?" column of the BES papers spreadsheet BEFORE you start collating any data. Go back and untick any you don't finish.

Take 10 papers to tackle at a time. Please don't take more papers until you have done these 10.

4.2 Finding your papers

Open the BES papers spreadsheet, find your first paper, then click the DOI link to the paper to view the online version. If you do not have access to the required sections of the paper, skip it and make a note of the paper number (do not enter any data for it) in the #skipped-papers Discord channel.

There is a Google Form to standardise the data collection. Please fill this in carefully. See details below. **Please especially check the DOI you have entered is correct** as we will use this to identify the paper throughout the analyses. And ensure the DOI you enter matches the paper you are adding data for. If you find any questions that are worded ambiguously, or you don't understand them, please let one of the organisers know so we can make it clear.

After you have got the hang of the data collection, it would be really helpful to know roughly how long it takes to enter data for one paper, so we can plan future data collection exercises. You can put this information into the Discord we'll use to communicate during the hack-a-thon.

5. Google form questions

Note that the form has several sections. Not all sections will appear for each paper. For example, if the paper has no data, it will automatically skip the data section. There are 6 sections in total.

If you cannot find the option you need in a dropdown menu/tick boxes please enter this as **Other** and **specify the option**. But also make a note of this and tell the organisers so we can add these options to the form. Use the #issues or #options-to-add

If you **cannot work out what to do for a paper, or it has an excessive amount of code that you don't feel you have the expertise to handle**, skip the paper, do not add any data for it, and note the paper number in the #skipped-papers Discord channel along with an explanation. You may also want to post a query in the #issues channel.

Section 1: GENERAL INFORMATION

1. **Paper number**

The number of the paper from the spreadsheet

2. **DOI of paper**

Please copy paste this from the spreadsheet and double check that you haven't made any errors

3. **Year of publication**

2015 - 2024. Please use the year listed on the spreadsheet.

4. **Journal**

This is listed on the spreadsheet.

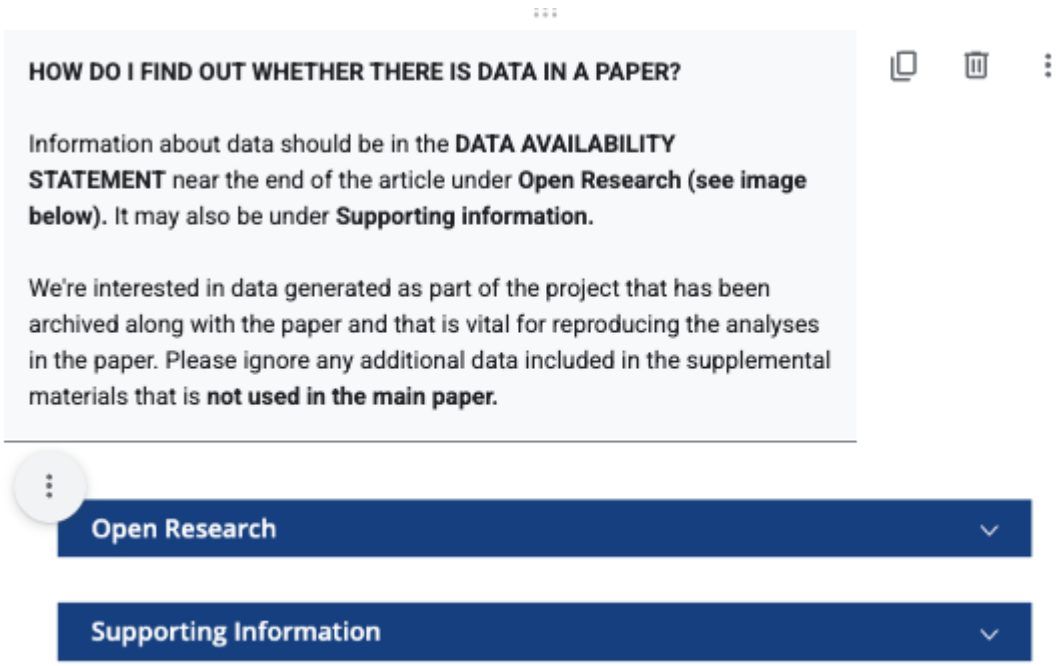
5. **Article type (on spreadsheet)**

This is listed on the spreadsheet.

Options are **Research Article**, **Application**, **Practical Tools**, **Long Term Study** and **Data Article**.

If there is a mismatch between the paper you are looking at online and details on the spreadsheet, check that you are looking at the correct paper. If there is still a mismatch let one of the organisers know via the #issues Discord channel and skip the paper in question. Do not add any data for it, and note the paper number in the #skipped-papers channel.

For clarity the following is shown before you are asked the next question:



6. **Please paste the text of the Data Availability statement here**

This is just in case we want to do any follow up with these data. Copy paste from the paper.

7. **Does the paper have data?**

Yes/No.

If the answer is Yes you will be taken to the Data and Data Equity sections.

Note that we are only interested in data mentioned in the data availability statement, or supplementary materials, and only if it is generated as part of the project and is vital to reproducing the results of the paper. I.e. some people will include data that is not relevant to the paper, just additional information. We don't care about this data.

WE ARE INTERESTED IN DATA WHERE IT IS A COLLATION OF EXISTING DATASETS. This data would still need to be archived as it is needed to reproduce the analyses. So please choose YES here.

Section 2: DATA EQUITY

We're interested in understanding if data use (and reuse) is equitable. These questions aim to identify whether data is being collected from places the authors are based in or if helicopter/parachute science is occurring. **FOR THESE QUESTIONS ONLY MENTION DATA THAT HAS BEEN COLLECTED IN THE LAB OR FIELD - ANALYSES WHERE PEOPLE USE DATA FROM PREVIOUS STUDIES OR COLLECTED ONLINE DO NOT COUNT HERE.**

This information should be easily available from the abstract or title of the paper. If you can't find out quickly choose Unclear.

1. **Country of corresponding author**

This is listed on the spreadsheet.

Dropdown menu. Select from UN M49 standard georegions:
https://unstats.un.org/unsd/methodology/m49/#geo-regions

Note that this list is georegions so there may be a more detailed descriptor of the region than simply the country. For example, islands have their own descriptor, rather than coming under the name of the country that they formally belong to.

2. **Country of first author**

Use the country on the first author's main address on the paper. There may be more than one. We mean the address that is next to their name on the paper, NOT any "current address" that may be added for people who have recently moved institutions.

Dropdown menu. Select from UN M49 standard georegions:
https://unstats.un.org/unsd/methodology/m49/#geo-regions

Note that this list is georegions so there may be a more detailed descriptor of the region than simply the country. For example, islands have their own descriptor, rather than coming under the name of the country that they formally belong to.

3. **What georegion(s) were any novel data used in the paper collected from?**

NOVEL here means new data collected for the purposes of this study. So if it uses stuff from previous studies that has been collated into a larger dataset but none of the data were physically collected for that paper, no novel data have been collected (but this still counts as data we would want to be archived for the other questions).

Select from UN M49 standard georegions:
https://unstats.un.org/unsd/methodology/m49/#geo-regions. Apologies there are a lot of options!

Note that this list is georegions so there may be a more detailed descriptor of the region than simply the country. For example, islands have their own descriptor, rather than coming under the name of the country that they formally belong to.

4. **Are any authors based in the georegion(s) the data were collected from?**

Use their main address on the paper. i.e. if the data were collected in Afghanistan, do any authors have an address in Afghanistan?

Also check the **Acknowledgements** to see if data labourers from the georegion who are not authors are mentioned.

Options: (you can select more than one)

- No authors are based in the georegion the data came from
- Yes - First author
- Yes - Last author
- Yes - One or more middle authors
- Yes - Data labourers from the georegion are in the Acknowledgements
- Data labourers are in the Acknowledgements but it is unclear where they are based.
- Unclear
- Other

Section 3: DATA

We're interested in data generated as part of the project that has been archived along with the paper and that is vital for reproducing the analyses in the paper. Please ignore any additional data included in the supplemental materials that is not used in the main paper.

1. **Are the data mentioned in the Data Availability statement?**

Yes/No

2. **Are you able to find the data using the link/instructions in the Data Availability statement?**

Yes/No

3. **Where are the data archived (select all that apply)?**

Options: (you can choose more than one option)
- DataVerse
- Dryad
- Figshare
- GBIF
- GitHub, GitLab, Codeberg or similar platform
- Institutional repository
- NCBI GenBank
- OSF
- Personal website
- Supplementary materials
- Zenodo
- Other (please specify)

4. **Does the dataset have a DOI?**

Yes/No/Unsure/Other (please specify)

5. **Can you download the data?**

Yes/No

6. **Can you open the data?**

Yes/No/Maybe if I had the correct software/Other (please specify)

7. **What format are the data in?**

Options: (you can choose more than one option)

- .csv/tsv
- .doc(x)
- .nex
- .odt
- .ods
- .pdf
- .sql(ite)
- .txt
- .xls(x)
- Other (please specify)

8. **Does the data have a README/metadata?**

Yes/No/Unsure/Other (please specify)

9. **How useful is the README/metadata?**

A good data README should include

- Information on the manuscript it came from.
- Contact details of at least one author.
- License information.
- Information about the data
    - a brief summary of how data were collected,
    - Sources of data if from a literature review
    - List of all data files
    - Column-by-column description of the data files, along with column headers, measurement units, levels of factors (e.g. if the variable is “habitat” what categories are possible?), explanations for any abbreviations.
    - See Data Dictionary explanation here.

Scale of 1-10
1 = Very brief and incomplete.
10 = You can understand the dataset in just a few minutes. Contains all column headers, abbreviations, units, data sources, data dictionary, license info, manuscript info etc.

10. **How complete are the archived data?**

Archived data should contain **all the data necessary to reproduce all the analyses and results**. Issues of completeness include providing only summary data, not providing raw data, providing the wrong data, and providing insufficient data to repeat the analyses. This might be a bit difficult to assess but let’s give it a go!

Options:

- The main analyses of the paper cannot be repeated with the data that has been archived.
- Some analyses can be repeated but not all (~50% of analyses can be repeated).

- Most data are provided with only small omissions, for example exploratory analyses (~75% of analyses can be repeated).
- All the data necessary to reproduce **all** analyses and results are archived.
- Unsure
- Other (please specify)

11. Does the data have a license?

Yes/No/Unsure/Other (please specify)

12. If so, which license?

- CC BY
- CC BY-NC
- CC BY-ND
- CC BY-NC-ND
- CC BY-NC-SA
- CC BY-SA
- CC0
- Other (please specify)

13. Has the data been cited (excluding the original paper it was archived for)?

You can find this information on the Dryad and Figshare landing pages for the dataset on the right hand side. For other repositories it varies. If you can't find out easily just pick Unsure.

Yes/No/Unsure/Other (please specify)

14. If yes, how many times?

Enter number or “Unclear”

15. Has the data been cited by someone other than the original authors?

Yes/No

16. If yes, how many times?

Enter number or “Unclear”

17. For any paper(s) citing the data, what georegions are on the authors main addresses (i.e. when doing the work)? Please select all relevant options for all authors on the paper.

Select from UN M49 standard georegions: https://unstats.un.org/unsd/methodology/m49/#geo-regions. Apologies there are a lot of options! This is to tackle questions about data equity.

Note that this list is georegions so there may be a more detailed descriptor of the region than simply the country. For example, islands have their own descriptor, rather than coming under the name of the country that they formally belong to.

Section 4: CODE

We're interested in code used as part of the project that has been archived along with the paper and that is vital for reproducing the analyses in the paper. Please ignore any additional code included in the supplemental materials that is not used in the main paper.

Note that for MEE papers only, there is an additional spreadsheet compiled by Huijie which has GitHub links and programming language on it which may speed up data collection for the CODE section. You will have to search for the DOI to find the papers as these do not have paper numbers

Some of this may be really hard for papers with a lot of code, especially if you are not so comfortable with programming. If you get a paper with a lot of code feel free to a) ask for help on Discord; b) skip the paper (and note it in the #skipped-papers channel). This should not happen too frequently.

We first want to determine if papers are using code at all - many will use code but not archive it. Some will archive it but not mention this in the Data Availability Statement, instead referring to this in the Methods or Supplementary Materials.

Just adding a link to EOSC's guidelines for Research Software MetaData here in case people are interested. It's very dry, but an enormous amount of work went into it: doi:10.5281/zenodo.10786147. Their more readable site is here: https://fair-impact.github.io/RSMD-guidelines/

HOW DO I FIND OUT IF A PAPER USED CODE?

If code is archived (see next question) this is an easy "Yes", but if there is no obviously archived code, check the Methods to see if any common programming languages are mentioned. For example, if the authors mention doing analyses in R, they must have code somewhere even if it is not archived.

0.1. Does the paper use code?

Yes/No/Unsure/Other

0.2. If the paper uses code, and this is NOT archived (i.e. they mention using R but don't provide any scripts) please copy-paste the text from the paper that alerted you to them using code…*(just choose the first instance you notice, no need to include every mention).*

E.g. We used R version 4.5 in all analyses. OR We used the Python function "lambda" etc.

0.3 Does the paper have archived code?

Yes/No/Other

If the answer is Yes you will be taken to the Code section.

1. **Is the code archiving described in the Data Availability statement?**

Yes/No - but it is described in the Methods/No - but it is described in the Supplemental Materials/No/Other

***Please paste the section of the text that refers to the archived code if this is NOT part of the Data Availability Statement*.**

Paste in relevant text, e.g. All scripts are available from GitHub OR Code is available at [LINK]

2. **Are you able to find the code using the link/instructions in the Data Availability (or equivalent) statement?**

Yes/No

3. **Where is the code archived?**

Options (you can choose more than one option)

- DataVerse
- Dryad
- Figshare
- GitHub, GitLab, Codeberg or similar platform
- Institutional repository
- OSF
- Personal website
- Supplementary materials
- Zenodo
- Other

4. **Does the code have a DOI?**

Yes/No/Other

5. **Can you download the code?**

Yes/No/Other

6. **Can you open the code?**

Yes/Maybe if I had the right software/No/Other

7. **What language is the code written in?**

NOTE: This question and the next one may be hard for papers with a lot of code, so if you get a paper with a lot of code and this question and the next one are taking you a lot of time, feel free to skip these questions and make a note in the #skipped-papers channel that you skipped them. Do the rest of the paper if you can.

Options (you can choose more than one option)

- R
- Python
- Julia
- C/C++
- CSS
- Fortran
- Go
- HTML
- Java
- Markdown
- MATLAB
- Perl
- Rebol
- Rust
- Shell
- SPSS/PSPP
- SQL
- TeX/LaTeX
- Visual Basic
- Other

8. **What format is the code in?**

Note that the point of this is to work out if people are using appropriate formats, or are they storing R code in Word documents (eek!).

Options: (you can choose more than one option)

- .csv
- .doc(x)
- .md
- .ods
- .odt
- .pdf
- .py
- .qmd
- .R
- .RMd
- .sql
- .txt
- .xls(x)
- Other

9. **Does the code have a README?**

Yes/No/Other

10. **How useful is the README?**

A good code README should include…

- Information on the manuscript it came from.
- Contact details of at least one author.
- License information [note that some people provide this as a separate LICENSE file which is also good practice].
- List of all scripts and what they do, i.e. processing, analysis, plotting etc. and what their outputs are (e.g. table 1, figure 2). [note that some of the detailed descriptions of this may be in the files themselves, especially for functions, this is also fine but the README should list the basics of what the scripts do].
- Details of the workflow of the code if there are multiple scripts, i.e. what order do the scripts need to be run in?
- How does the code link to the data? i.e. which data files are needed for each script?
- The name of the software used (e.g. R), version, and names and versions of all packages required to run the analyses.

Note that for things like R packages READMEs may be a little different as other documentation may be included instead. Ask for help if you get a package to review but you're not sure if the documentation provided with it is sufficient.

Scale 1-10
1 = Very brief and incomplete.
10 = All information about script functionality, outputs, software, packages, workflows comprehensively documented.

11. **How well is the code annotated?**

Good code annotation includes:

- A header at the top of each script with a title and a quick overview of what the code does.
- Code should be divided into distinct sections with clear headings describing their purpose (e.g. loading packages, data processing, plotting, analysis etc.).
- Annotations should clearly describe what the code does, how to run it and what it produces.
- For code that takes a long time to run, comments about run times are also useful

1 = Not at all
10 = Thorough commenting throughout.

12. **Does the code have a separate vignette with examples of how the code should be used?**

Note this is most common for Applications papers where the code/software is the main output of the paper. It is not usual for papers that are just archiving the code used to run their analyses.

Yes/No/Other

13. **If the code is an R Package, and it was previously available on CRAN/Bioconductor (as stated in the Data Availability statement), is it still available?** Check this by trying to install it on the most recent version of R. If you don't know how to do this/can't do this choose "Unable to check"

Yes/No/Unable to check/Other

14. **If the code is a package in another language (e.g. Python published on PyPI) please put the name of the package here and we will see if it is still available**. If you know how to look this up yourself, please do and mention whether it is still available here…

Enter name of package and where it was deposited. Plus Yes/No if you know how to see if this is still available.

15. **Does the code have a CITATION file?**

Yes/No/Unclear/Other

16. **Does the code have a license?**

Yes/No

17. **If yes, what license?**

Options:

- AGPL
- Apache
- BSD variants
- CC BY
- CC BY-SA
- CC0
- EUPL
- GPL
- LGPL
- MIT
- MPL
- Unclear
- Other

18. **APPLICATIONS** papers only. How many times has the paper been cited ?

Enter number. This can be found on the webpage for the paper.
We have added this question because for packages people are more likely to cite the paper than the code itself.

19. **Has the code been cited (excluding the original paper it was archived for)?** Code = the archived code (not the paper or package).

Yes/No

20. **If yes, how many times?**

Enter number/unclear

21. **Has the code been cited by someone other than the original authors?**

Yes/No

22. **If yes, how many times?**

Enter number/unclear

Section 5: ADDITIONAL COMMENTS

1. **Did you have any issues recording data for this paper? If yes we will go back and check it for you later.**

Yes/No

2. **Any other comments about this paper?**

Enter text

Section 6: RECORDER INFORMATION

1. **Who recorded data from this paper? First name and initials only**

Enter your first name and initials (due to GDPR legislation) e.g. Natalie NC. If working in pairs, record both names separated by a comma.

2. **What date did you do this (YYYY-MM-DD)?**

Enter the date in YYYY-MM-DD format.

THANKS!!!